# Different Types of Nerve tissue-I (Neurons and Neuroglia)

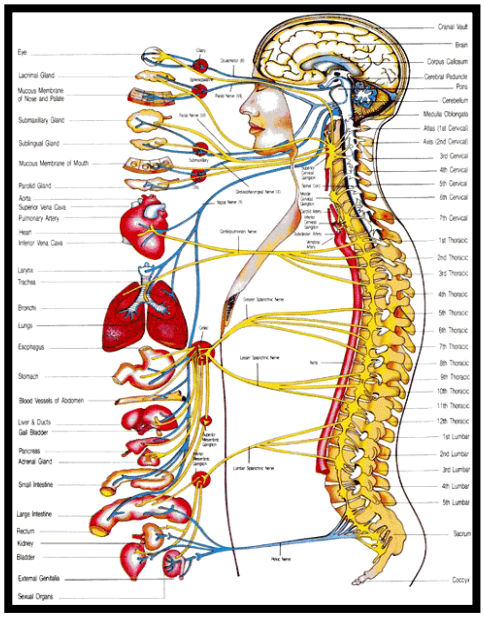
## Learning Objectives

**At the end of lecture, students should be able to:**

1. **Describe the nervous tissue**
2. **Define neuron , its structure and function**
3. **Enlist types of neurons according to structure**
4. **Enlist types of neurons according to functions**
5. **Define neuroglia, their types and functions**

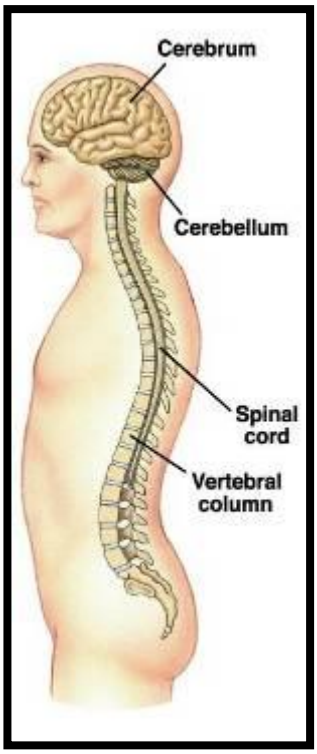
## NERVOUS SYSTEM

- Highly organized system of human body.
- It is the organ system containing a network of specialized cells called neurons that coordinate the actions of an animal and transmit signals between different parts of its body.



# ORGANIZATION OF THE NERVOUS SYSTEM

- Anatomical Divisions:
  1. Central Nervous System
     - The brain + the spinal cord
       - The center of integration and control
  2. Peripheral Nervous System
     - The nervous system outside of the brain and spinal cord Consists of:
       - 31 Spinal nerves Carry info to and from the spinal cord
       - 12 Cranial nerves Carry info to and from the brain




## ***BASIC FUNCTIONS OF THE NERVOUS SYSTEM***

### 1. Sensation:

- Monitors changes/events occurring in and outside the body. Such changes are known as stimuli and the cells that monitor them are receptors.

### 2. Integration:

- The parallel processing and interpretation of sensory information to determine the appropriate response

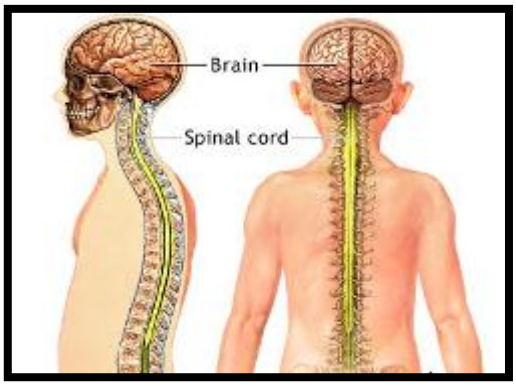
### 3. Reaction/Response:

- Motor output.
- response to information processed through stimulation of *effectors*
  - muscle contraction, glandular secretion

The activation of muscles or glands (typically via the release of neurotransmitters (NTs)

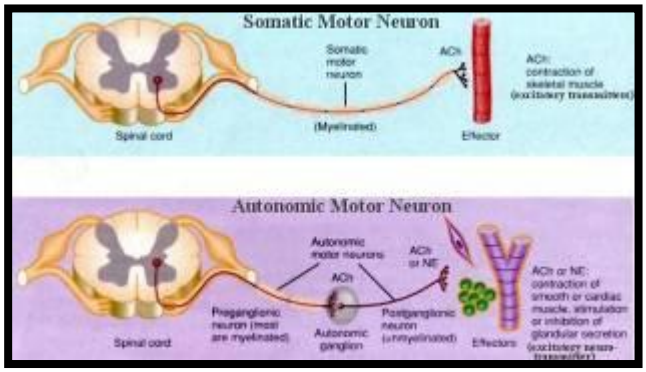
## ***CENTRAL NERVOUS SYSTEM***

- Divided into two parts:
- ***Central N.S:***
  - **Consists of**
  - **Brain,** contained within the cranium
  - **Spinal cord,** lodged in the vertebral canal
  - Two portions are continuous with one another at the level of the upper border of the atlas vertebra.



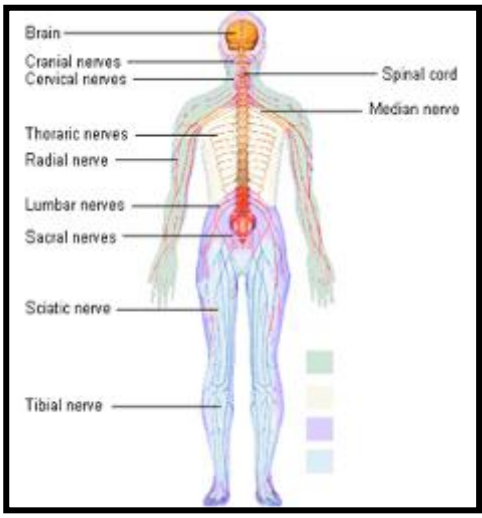

## ***Peripheral nervous system***

- ***Peripheral N.S:***
- It is a series of nerves by which the central nervous system is connected with the various tissues of the body.
- PNS is not protected by the bone of spine and skull, or by the blood-brain barrier, leaving it exposed to mechanical injuries.
- Divided into:
  - Somatic nervous system
  - Autonomic nervous system



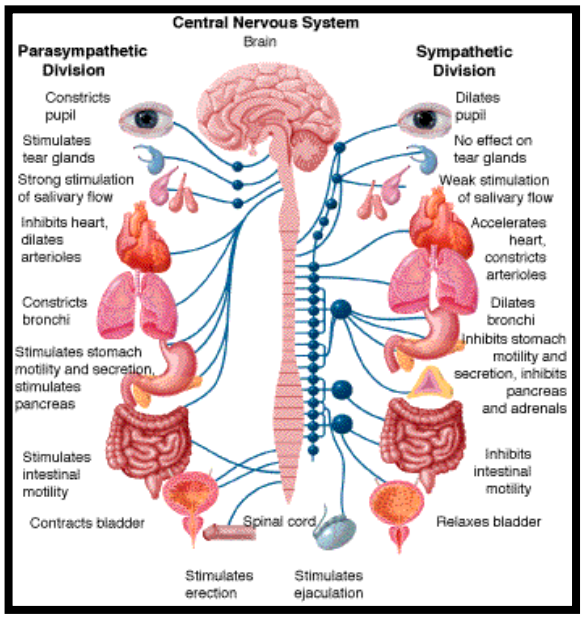
## ***SOMATIC NERVOUS SYSTEM***

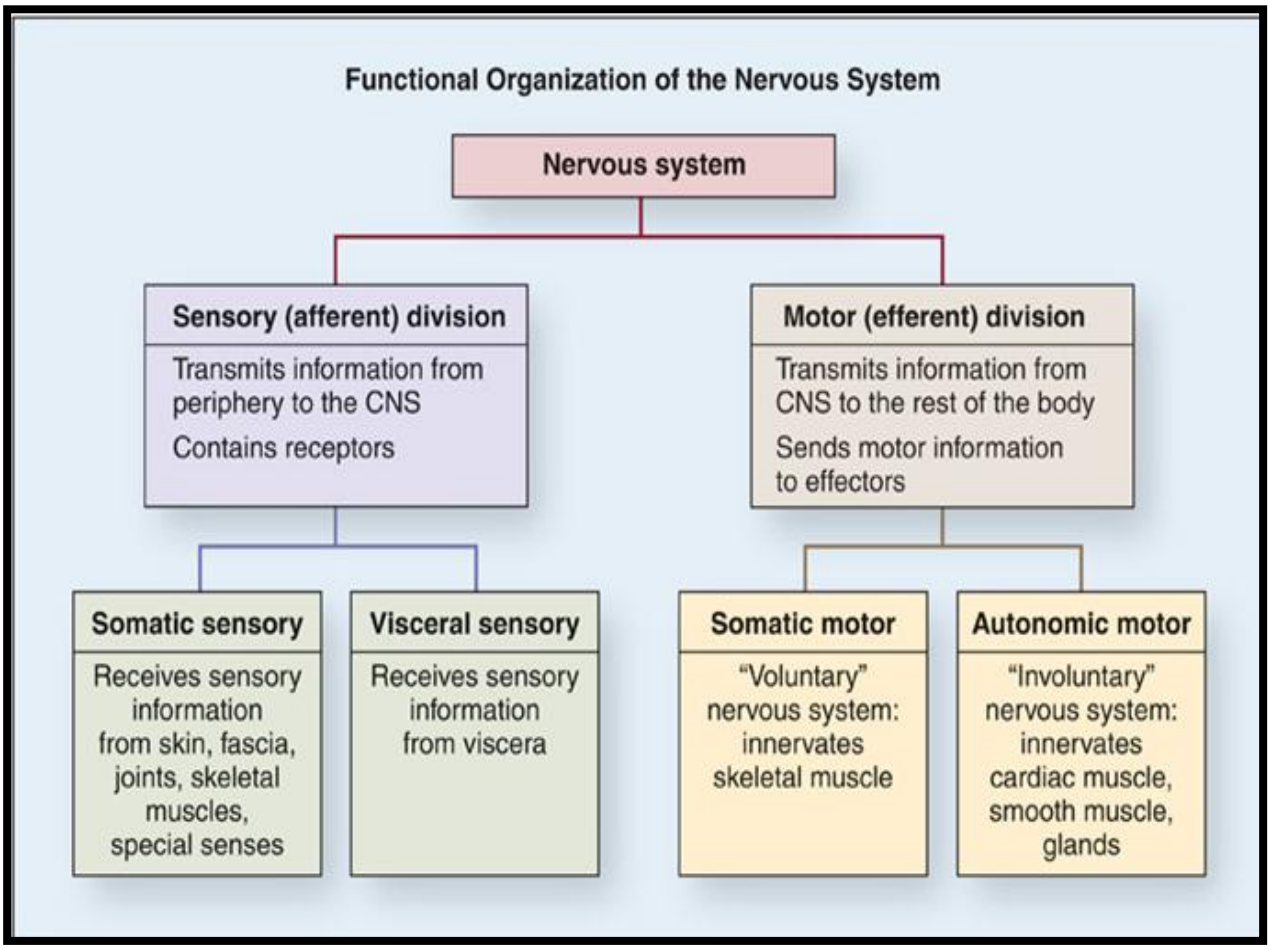
- Associated with the voluntary control of body movements via skeletal muscles, and sensory reception of external stimuli (e.g., touch, hearing, and sight)
- Cerebrospinal nerves are 43 in number on either side
- 12 **cranial,** attached to the brain
- 31 **spinal,** attached to the spinal cord.




## ***AUTONOMIC NERVOUS SYSTEM***

- The ANS affects heart rate, digestion, respiration rate, salivation, perspiration, diameter of the pupils, micturition (urination), and sexual arousal.
- Whereas most of its actions are involuntary, some, such as breathing, work in tandem with the conscious mind.






# Nervous Tissue

- Nervous Tissue comprises of
  Neurons,
  Neuroglia,
  and blood vessels

- **Neuron** is the basic structural and functional unit of nervous tissue

- **Neuroglia** is the supporting or glial tissue that is 10-times more abundant in mammalian brain than neurons

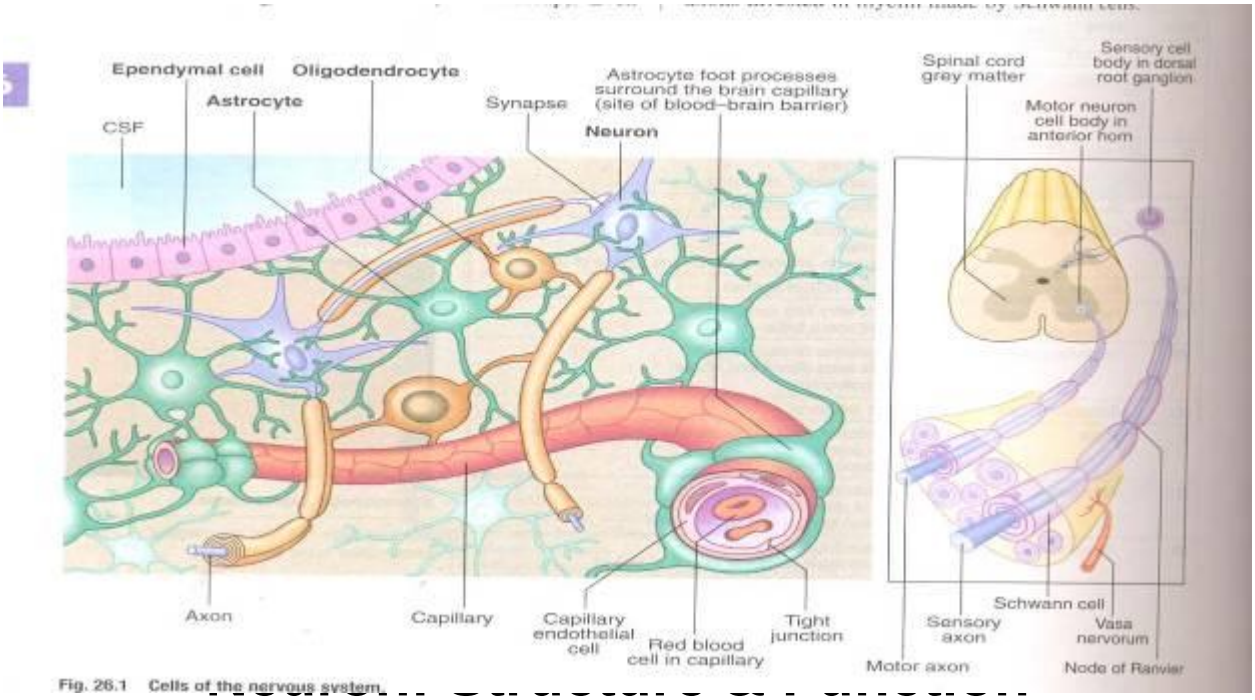


Fig. 26.1 Cells of the nervous system.

## Parts of Neuron

**Body:** Soma/Perikaryon

*Body Contains;*

Nucleus:

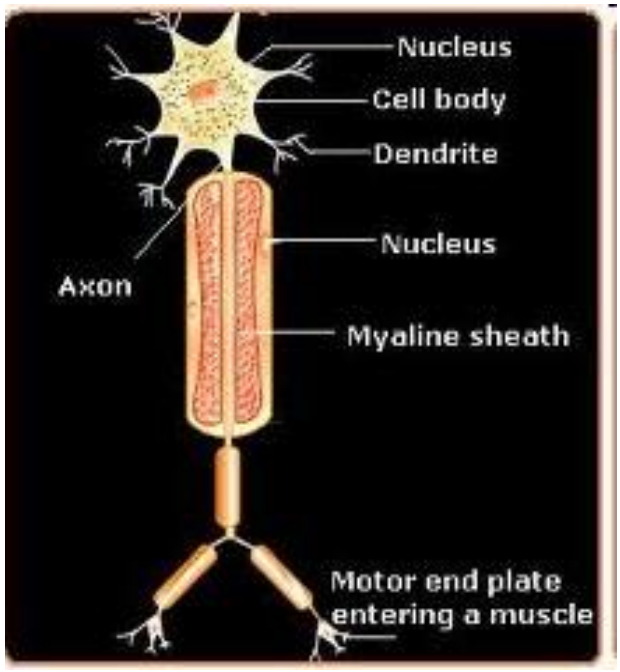
Cytoplasmic organelles

Inclusions

Cytoskeletal components

**Processes**:

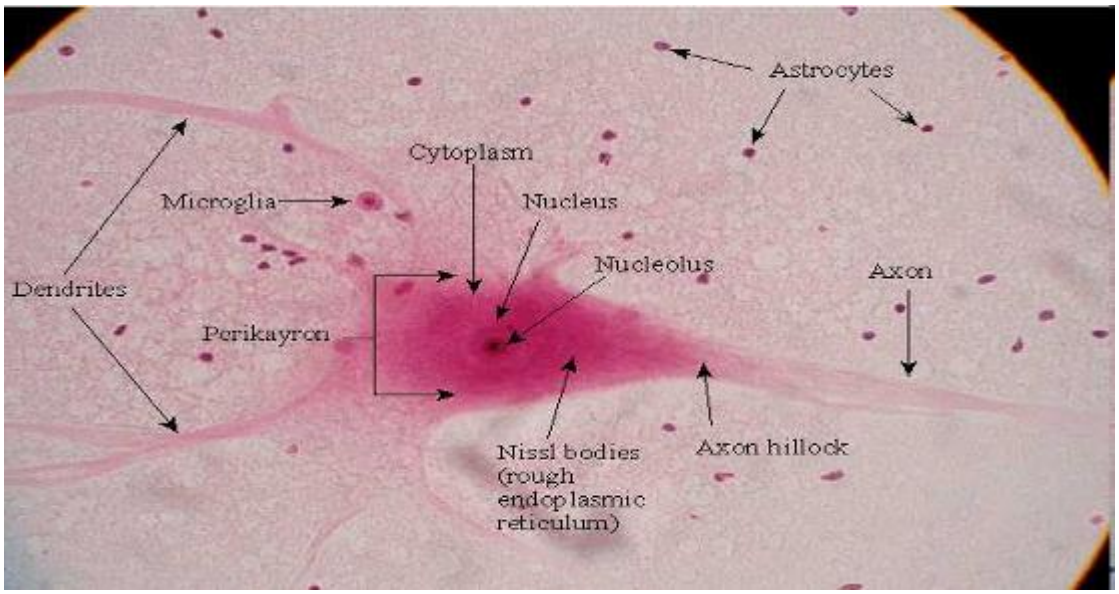
Axon and Dendrites



# Neuron: Structure & Function

## Nucleus:

- Centrally located
- Large
- Pale-staining
- Spherical (Owl-eye appearance, large nucleolus)
- Euchromatin (shows intense activity of cells)
- Bi-nuclear neurons in sensory and symp-ganglia



## Cytoplasmic Organelles

- Nissel bodies
  - Polysomes and RER
  - More abundant in motor neurons
- Golgi complex (located near nucleus)
- Mitochondria (scattered diffusely cytoplasm)
- No centriole (non-capable of division)

## Inclusions

- Melanin containing granule in CNS neuron and dorsal root sympathetic ganglia
- Lipofuscin granules in some neurons increase with age
- Lipid droplet present occasionally

## Cytoskeletal Components

- Neurofilaments
- Microtubules
- Microfilaments

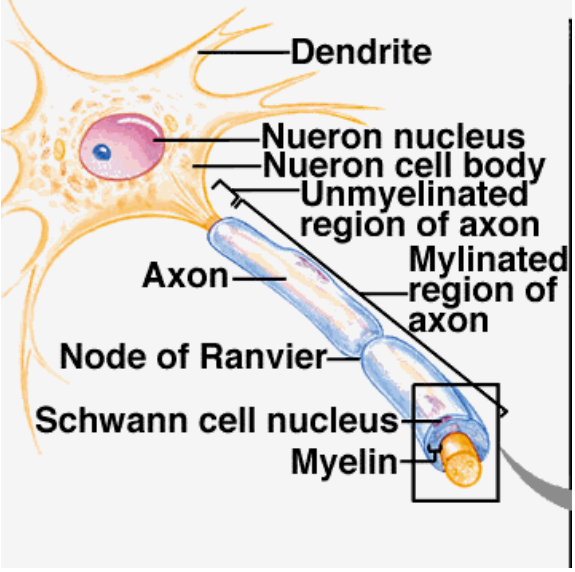
## Processes: Dendrites

- Input receiving process
- Conveys afferents signals to Soma
- Usually multiple and small in length
- Thickness decreases near terminals
- Arborized terminals
  - except bi-polar neuron
- Cytoplasmic organelles similar with Soma
  - only lack golgi complex
- Organelles reduces near terminals

– except mitochondria

## Processes: Axons

- Output receiving process
- Conduct impulse away from soma
  - … to axon terminals – efferent signals
- Different lengths, golgi type-I & type-II
- Single and usually long
- Diameter remains uniform through length
- Originates from axon hillock
  - Specialized region of soma
  - lacks RER, ribosomes
  - shows many microtubules & neurofilaments

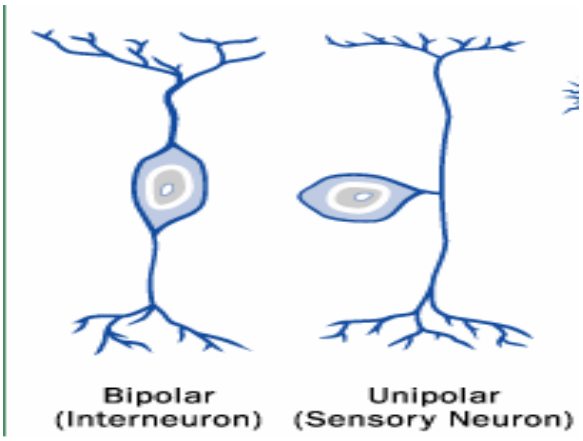


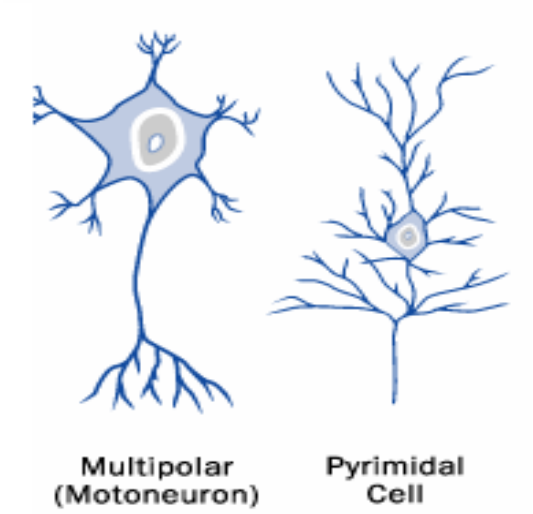

## Neuron: Morphological Classification

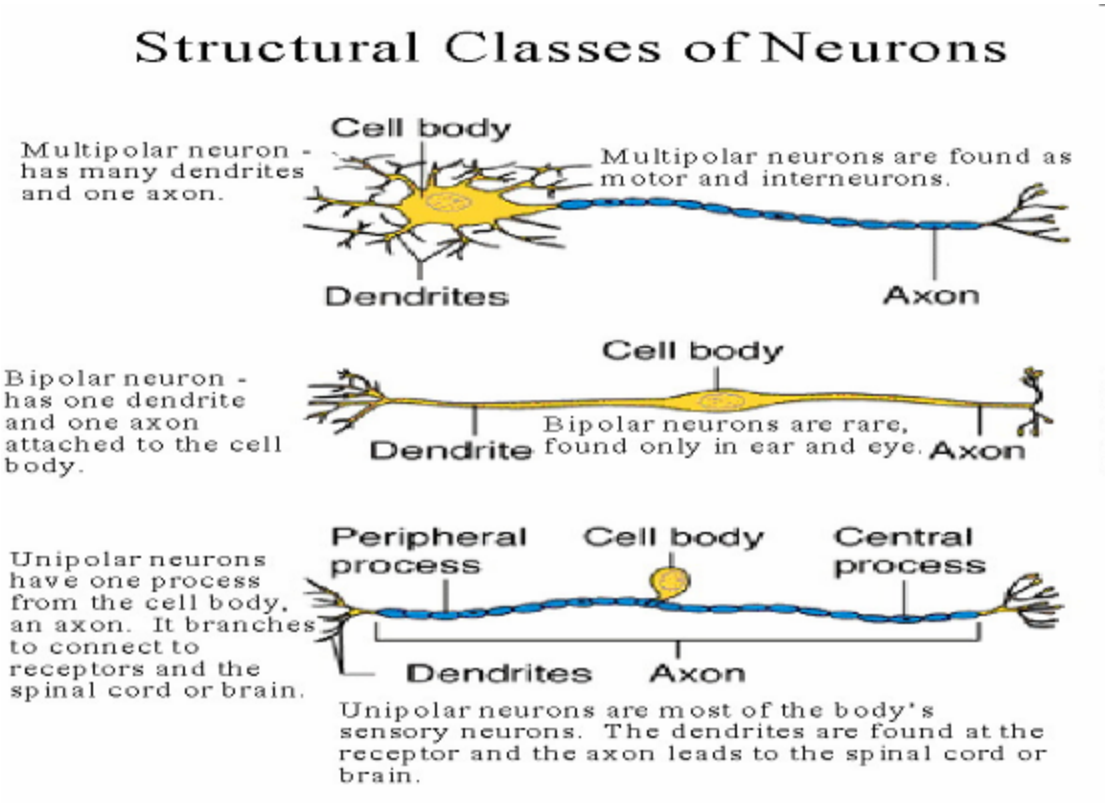
- **Unipolar**
- **Bipolar**
- **Multipolar**

- Single axon & multiple dendrites
- Most common type in men, e.g
- Motor cortex, interneurons, …

- **<u>Pseudounipolar</u>**
  - Single process arises from body
  - Branches into an axon and dendrite, e.g
  - Present in spinal and cranial ganglia






# Neuron:Functional Classification

- **Sensory/Afferent: 3-types of receptions**
  - Exteroceptors (external: touch, temp)
  - Proprioceptors (position: muscle & joints)
  - Interoceptors (internal: GIT, UGT, RT)
- **Motor/Efferent: 2-types of commands**
  - Somatic motor (PNS-skeletal muscles)
  - Visceral motor (ANS-gut)
    - Pre-ganglionic fiber & Post-ganglionic fiber
- **Interneurons**
  - Runs b/w sensory & motor neurons
  - Most numerous type – only in CNS
  - Coordinates b/w sensory inputs & motor outputs

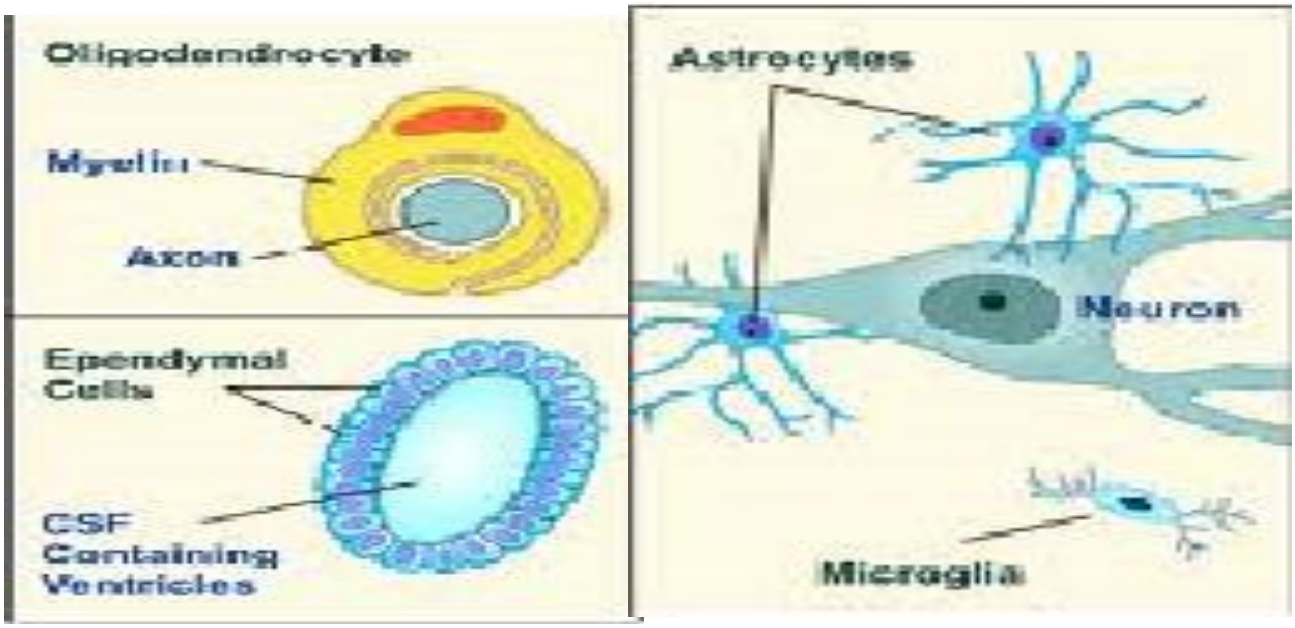
# CNS Neuroglia / Neuroglial tissue

– **Astrocytes(macroglia)**
  - **protoplasmic & fibrous**
  - **Helps establish & maintain blood-brain barrier**

– **Oligodendrocytes (macroglia)**
  - **Produce myelin**
    – **a sheath that wraps axons of several cells**

– **Microglia (phagocytic cells)**
  - **Several cells with multiple branching processes**

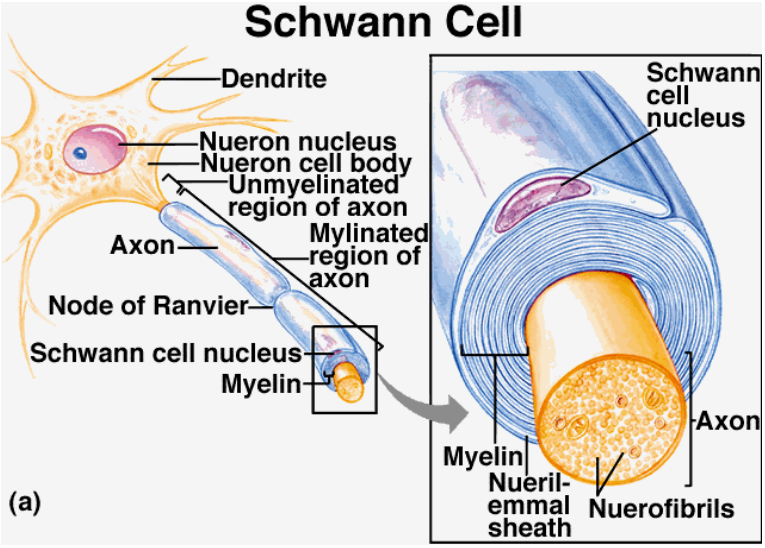
– **Ependyma (epithelial cells with cilia)**
  - **Contributes for choroids plexus formation**




# PNS Neuroglia

- **Satellite Cells**
  - Found in ganglia, around nerve cell body
  - Functions as exchange of nutrients and waste

- **Schwann Cells (neurolemmocytes)**
  - Flattened cell produce myelin sheath in PNS
    - … that surround axons
    - … only associated with a single axon
  - Forms myelinated or unmyelinated coverings over neurons




# Neuroglia vs. Neurons

- **Neuron** is the basic structural and functional unit of nervous tissue

- **Neuroglia** is the supporting or glial tissue

- Neuroglia divide.
- Neurons do not.
- Most brain tumors are “gliomas.”
- Most brain tumors involve the neuroglia cells, not the neurons.